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PART A

Task 1

1. Cleanup rule
2.

No.	Name	Source	Destination	VPN	Services & Applications
1		None	None	Any	None
2	Cleanup rule	Any	Any	Any	Any

Figure 1a Cleanup rule

Action	Track	Install On
Drop	None	Policy...
Accept	Log	Policy...

Figure 1b Cleanup rule

3.

No.	Hits	Name	Source
1	0		None
2	405	Cleanup rule	Any

Figure 2a

No.	Hits	Name
1	0	
2	405	Cleanup rule

Figure 2b

4. The three categories of settings under Global Properties are Open Object Explorer, Publish Session and Install Database. Alpha Standard and Standard are the policy packages that exist under Manage Policies and Layers.

Task 2

No.	Name	Source	Destination	VPN	Services & Applications
1	Admin access	Alpha-Admins	A-GW-Cluster	Any	ssh https
2	Stealth rule	Any	A-GW-Cluster	Any	Any
3	Inbound to DMZ	Any	Alpha-Web	Any	http https smtp
4	Internal to DMZ	None	None	Any	None
5	DNS out	InternalNet	ExternalZone	Any	dns
6	Internal web out	InternalNet	Any	Any	http https
7	SMTP out	Alpha-Web	Any	Any	smtp
8	Cleanup rule	Any	Any	Any	Any

Figure 3a

Action	Track	Install On
Accept	Log	Policy...
Drop	Log	Policy...
Drop	None	Policy...
Drop	None	Policy...
Accept	Log	Policy...
Accept	Log	Policy...
Accept	Log	Policy...
Drop	Log	Policy...

Figure 3b

Task 4 Organize the rulebase with sections

No.	Name	Source	Destination	VPN	Services & Applicatio
Management and Protection (1-2)					
1	Admin access	Alpha-Admins	A-GW-Cluster	Any	ssh https
2	Stealth rule	Any	A-GW-Cluster	Any	Any
DMZ Services (3-4)					
3	Inbound to DMZ	Any	Alpha-Web	Any	http https smtp
4	Internal to DMZ	None	None	Any	None
Outbound Services (5-7)					
5	DNS out	InternalNet	ExternalZone	Any	dns
6	Internal web out	InternalNet	Any	Any	http https
7	SMTP out	Alpha-Web	Any	Any	smtp
Cleanup (8)					
8	Cleanup rule	Any	Any	Any	Any

Figure 4a Rulebase with sections (all sections expanded)

No.	Name	Source	Destination	VPN	Services & Applications
Management and Protection (1-2)					
1	Admin access	Alpha-Admins	A-GW-Cluster	Any	ssh https
2	Stealth rule	Any	A-GW-Cluster	Any	Any
DMZ Services (3-4)					
3	Inbound to DMZ	Any	Alpha-Web	Any	http https smtp
4	Internal to DMZ	None	None	Any	None
Outbound Services (5-7)					
5	DNS out	InternalNet	ExternalZone	Any	dns
6	Internal web out	InternalNet	Any	Any	http https
7	SMTP out	Alpha-Web	Any	Any	smtp
Cleanup (8)					

Figure 4b Rulebase with sections (one collapsed section)

No. Section titles do not affect rule enforcement

Task 6 Configuration of Hide NAT

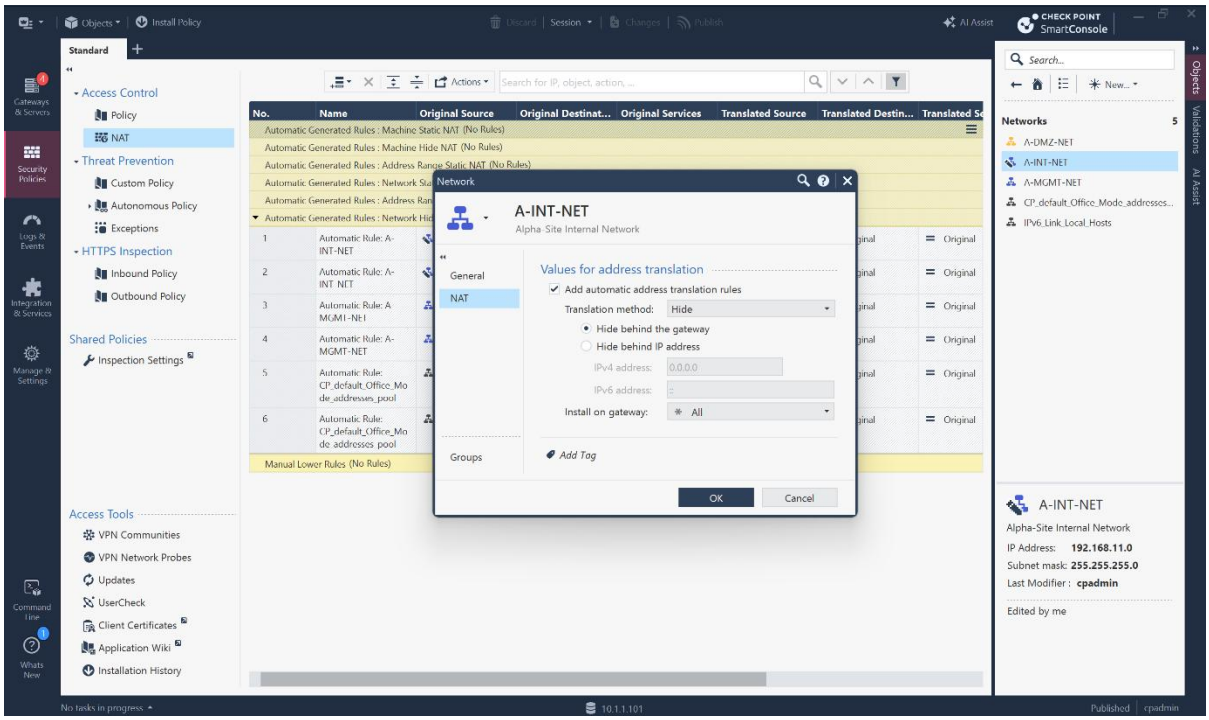


Figure 5a. Internal NAT Settings

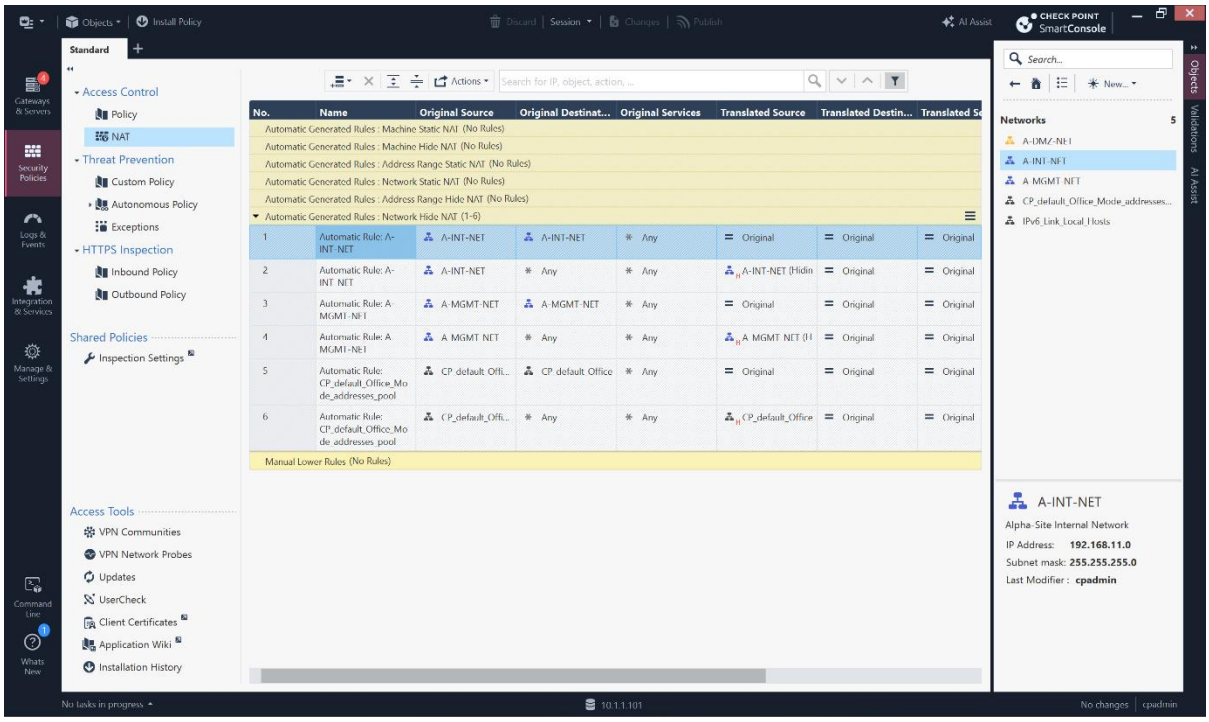


Figure 5b. NAT Rulebase

In Check Point SmartConsole, an **Automatic NAT rule** is generated and managed dynamically directly from within the network object's properties (under the NAT tab), rather than being explicitly written as a custom row in the manual NAT rule base.
When you select "Add automatic address translation rules" on the Internal_LAN object:

The Security Management Server automatically creates and enforces the translation logic behind the scenes whenever that object is used in policies.

Many-to-one translation (commonly referred to as Hide NAT) describes an architectural mapping where an entire pool or subnet of private internal IP addresses - the **"many"** (e.g., the dozens or hundreds of workstations on the internal network) are translated to share a single public IP address - the **"one"** (the external interface IP of the Security Gateway or a dedicated public IP pool).
This allows an entire organization to access external networks simultaneously while conserving scarce public IPv4 address space.

Because multiple internal hosts are all hiding behind the *exact same* single public source IP address when communicating with external servers, the Security Gateway must implement Port Address Translation to prevent session collisions.

Task 7 Rule order experiment and installation History

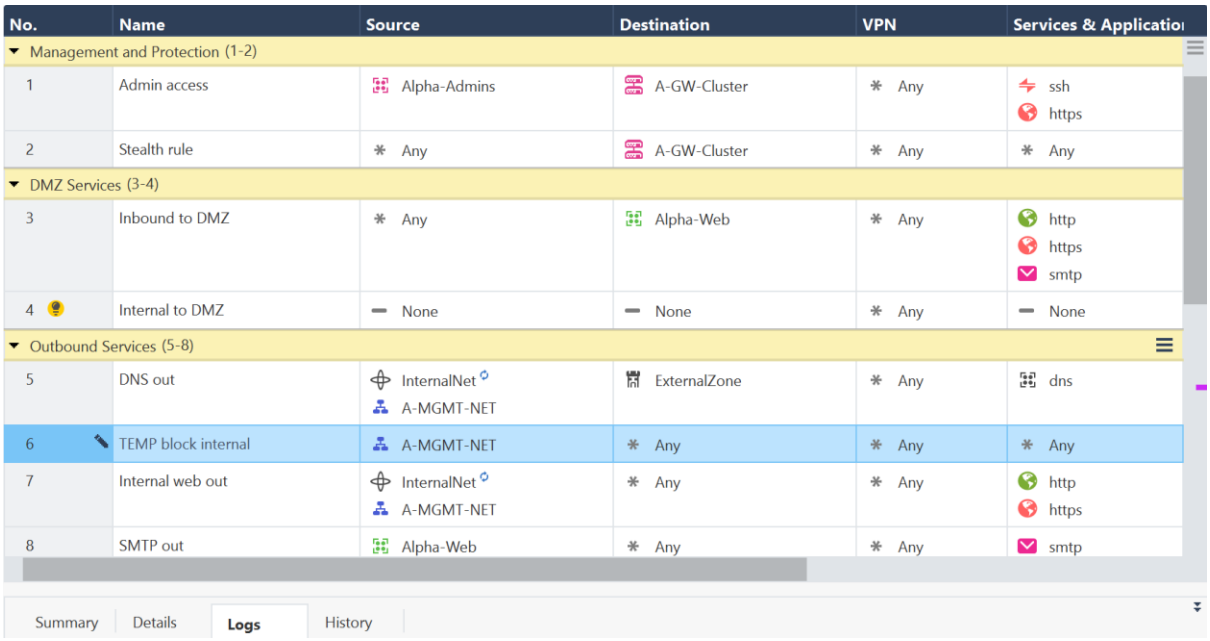
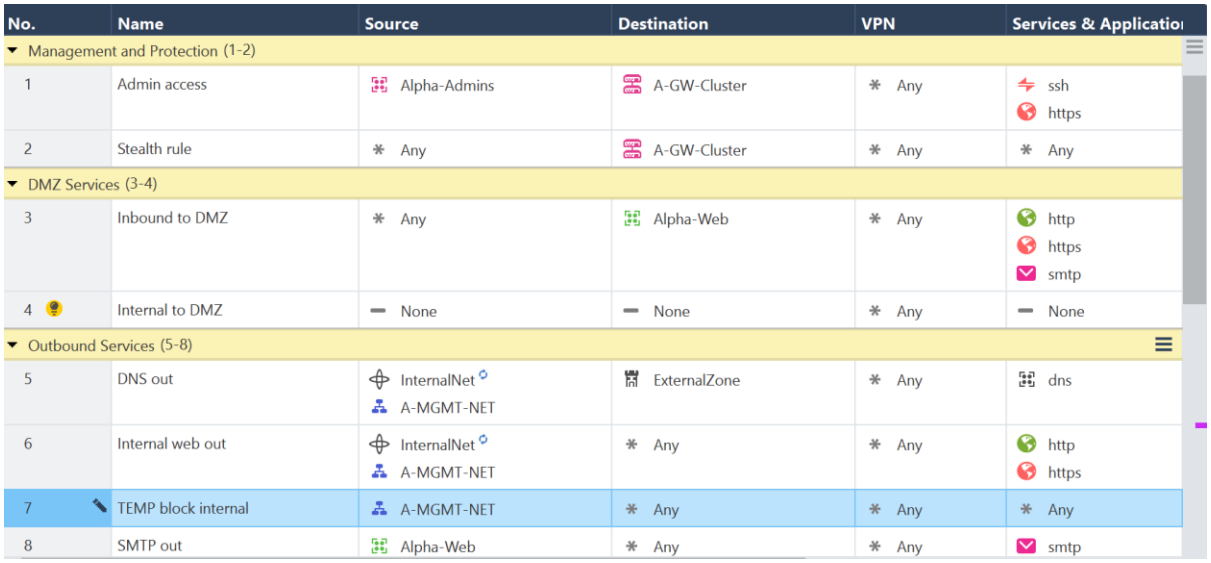


Figure 6a Rule order experiment and installation History



No.	Name	Source	Destination	VPN	Services & Applicatio
Management and Protection (1-2)					
1	Admin access	Alpha-Admins	A-GW-Cluster	Any	ssh https
2	Stealth rule	Any	A-GW-Cluster	Any	Any
DMZ Services (3-4)					
3	Inbound to DMZ	Any	Alpha-Web	Any	http https smtp
4	Internal to DMZ	None	None	Any	None
Outbound Services (5-8)					
5	DNS out	InternalNet A-MGMT-NET	ExternalZone	Any	dns
6	Internal web out	InternalNet A-MGMT-NET	Any	Any	http https
7	TEMP block internal	A-MGMT-NET	Any	Any	Any
8	SMTP out	Alpha-Web	Any	Any	smtp

Figure 6b Rule order experiment and installation History

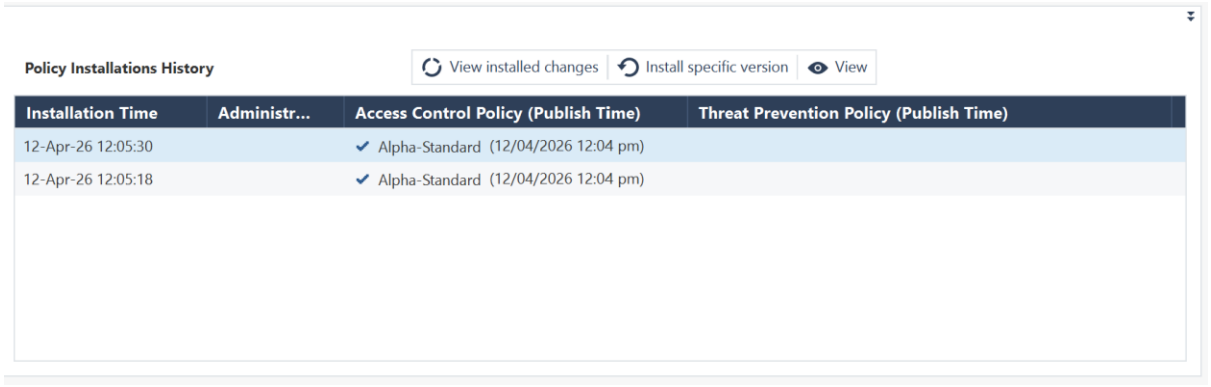


Figure 6c Publish and Install failed because of unavailability of Gateway

PART B

Section A

1. B
2. A
3. C
4. D
5. B

Section B2

6. **Explicit Rules vs. Implied Rules**

Rule Type	Definition & Control	Purpose
Explicit Rules	Manually created, edited and positioned by the administrator in the SmartConsole Rule Base	Implements organization-specific security policies and business requirements (e.g., allowing specific web, mail, or internal traffic)
Implied Rules	Built-in system rules automatically generated by Check Point to handle core administrative and control plane traffic	Maintains critical system communication – such as Secure Internal Communication (SIC), cluster heartbeats and management connections – without requiring manual rules

Table 1 Difference between Explicit and Implicit rules

Configuration and Placement of Implied Rules

- Where they are configured**:** Implied rules are managed globally in SmartConsole by navigating to **Menu > Global Properties > Firewall > Implied Rules**.
- Where they can be placed: place within **Global Properties**.

The Implicit Cleanup Action

The Implicit Cleanup Action is the built-in default behaviour executed by a policy layer when traffic matches none of the preceding explicit rules.

7. Security Best Practice dictates placing management protection rules – such as the Stealth Rule (which blocks and logs unauthorized access to the Security Gateway itself)—at the top of the explicit rule base.
   Placing the Stealth rule early ensures that external or untrusted traffic cannot attempt management logins or unauthorized connections before reaching user-defined traffic rules.

8. **Difference between saving and publishing**

Saving	Publishing
When you make changes in SmartConsole (such as creating a network object or modifying an access rule), those changes are saved **only to your private administrative session**	When you click **Publish**, all modifications in your current session are committed to the **main** Security Management Server database
Other administrators logged into the Security Management Server cannot see your updates, and your changes have zero impact on the global database	our session changes become permanent, creating a new audit log entry and a published database version. Once published, your modifications are visible to all other administrators

Table 2. Difference between Saving and Publishing

Unpublished changes are NEVER compiled or sent to the Security Gateway. They remain isolated in their local session draft and will not affect active traffic enforcement.

Organizations enforce strict Change Management protocols around the **Publish** operation for several critical reasons like database locking & reversion boundaries, concurrent administration conflicts, prevention of untested policy compilation and regulatory & audit compliance.

9. **Global Properties vs Policy Packages**

Feature	**Global Properties**	**Policy Packages**
Scope & Scope Level	**System-Wide (Global).** Applies universally to the Security Management Server and all Security Gateways managed under it	**Targeted / Specific.** Applies only to the specific Security Gateway(s) to which the package is explicitly assigned and installed
Function	Defines default gateway behaviours, system-wide operational parameters, and implied rules	Defines the exact operational security policy layers (Access Control, NAT, Threat Prevention) enforced on a gateway
Granularity	Uniform configuration across all firewalls (cannot be varied	Flexible per-gateway configuration (e.g., DMZ gateway vs. Branch gateway

Feature	Global Properties	Policy Packages
	per gateway without local overrides)	can have different policy packages)

Table 3. Global Properties vs. Policy Packages

Example of What Each Controls

Global Properties Example: Controls Implied Rules (such as automatically permitting Secure Internal Communication [SIC], ICMP Echo Requests, or management access before or after explicit rules) and default logging behaviour across all gateways.

Policy Package Example: Controls the specific Access Control Rulebase (explicit allow/drop rules, user access control, and NAT rules) and Threat Prevention profiles assigned to a particular enforcement point.

How to Access Each in SmartConsole

Opening Global Properties:

- Click on the **Menu** button (top-left corner of SmartConsole).
- Select Global Properties.

Opening Policy Packages:

- Click on the **Security Policies** tab in the left-hand navigation pane.
- Click on the active **Policy Package name** at the top of the policy view and select **Manage policies and layers**

10. The four predefined security zones

- Wireless zone
- External zone
- DMZ zone
- Internal zone

Rules for Assigning Gateway Interfaces to Zones

- **One Zone Per Interface:** A single Security Gateway interface can belong to **only one** Security Zone at a time.
- **Multiple Interfaces Per Zone:** Separate interfaces connected to entirely different physical or logical subnets can be assigned to the **same** Security Zone

Benefit of Using Zone-Based Rules

- **Policy Simplification and Maintainability:** Security zones allow you to write abstract, macro-level rules based on functional trust boundaries rather than static IP subnets.

11. Two Security Problems Caused by This Configuration

Placing this type of rule above the Stealth Rule creates severe operational and security vulnerabilities:

- **Complete Bypass of Security Inspection (Shadowing):** Because Check Point processes rules sequentially from top to bottom on a **first-match** basis, every single packet hitting the firewall will match this Any-Any-Any Accept rule. Every single rule placed below it—including user access restrictions, DMZ isolation, and threat prevention filters—is rendered completely useless (**shadowed**).
- **Exposure of the Gateway to Direct Attack:** The primary purpose of a **Stealth Rule** (which blocks and logs all direct traffic to the Security Management Server and Gateway interfaces) is to shield the firewall infrastructure from reconnaissance, unauthorized administrative logins, and exploit attempts. Placing an open Accept rule above it leaves the Security Gateway itself fully accessible to unauthorized connections across all ports and management services (such as SSH, HTTPS, or WebUI).

How to Correct the Mistake

- **Delete the rule entirely** if it was created by mistake or
- **Move the rule below the Stealth Rule** and immediately restrict its parameters (specify exact Source, Destination, and Service) so it reflects a legitimate, least-privilege business requirement instead of *Any*.

  Click **Publish** to commit the database update and then click **Install Policy** to push the corrected rule base to the Security Gateway.

Rules Allowed to Sit Above the Stealth Rule

The only rule type permitted (and standard practice) to be placed above the Stealth Rule is an **Explicit Management / Admin Access Rule**.

Why it is allowed: The Stealth Rule blocks all traffic directed to the gateway. If an explicit rule allowing authorized administrative hosts did not exist above the Stealth

Rule, legitimate administrators would be locked out from managing, monitoring or configuring the Security Gateway.

12. Two Benefits of NAT

- **IPv4 Address Conservation:** Network Address Translation allows an entire organization with hundreds or thousands of private internal IP addresses (e.g., RFC 1918 space) to share a single public IP address assigned to the Security Gateway's external interface.
- **Enhanced Network Security (Topology Hiding):** NAT masks the internal network structure and host IP addresses from external observers on the Internet, preventing direct reconnaissance and targeted inbound attacks against internal workstations.

Three Diagnostic Checks for Post-Configuration Internet Loss

- Verify Automatic NAT Rule and Hide-Behind Interface Selection

Inspect the generated NAT rule base in SmartConsole (or the network object's NAT configuration page) to ensure Automatic Hide NAT is enabled and correctly points to the external gateway interface or its primary public IP address.

- Review Access Control Policy for Outbound Traffic Matching

Check the Access Control rule base to verify that an explicit Accept rule exists permitting outbound traffic from the "Internal_LAN" to the Internet (and that it hasn't fallen through to the cleanup rule).

- Verify Gateway Default Route and Upstream Connectivity

Check the Security Gateway's routing table via the Gaia Portal or CLI to verify that a valid default route (0.0.0.0/0) exists pointing to the ISP next-hop gateway.